



USL Assignment 2

Implement the principal component analysis algorithm

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Dataset Description

Dataset: Cleveland Heart Disease Dataset

1. Dataset Source

The Cleveland Heart Disease dataset is obtained from the UCI Machine Learning Repository, a widely used academic database for machine learning research and experimentation.

2. Domain of Dataset

- Medical diagnosis
- Cardiovascular disease prediction
- Clinical data analysis

The dataset is used to analyze patient medical attributes for determining the presence of heart disease.



3. Dataset Size and Structure

- Total records: 303
- Records after cleaning: approximately 297
- Total attributes: 14
- Input features: 13
- Target variable: 1

Each row represents one patient's clinical examination record.

4. Description of Features

The dataset contains various medical and physiological measurements, including:

Demographic attributes

- Age of patient
- Gender

Clinical measurements

- Resting blood pressure
- Serum cholesterol level
- Fasting blood sugar

Diagnostic test results

- Resting electrocardiographic results
- Maximum heart rate achieved



- Exercise-induced angina

Cardiovascular indicators

- ST depression value
- Number of major vessels
- Thalassemia test result

The target variable indicates the presence or absence of heart disease.

5. Data Type Characteristics

- Mostly numerical attributes
- Some categorical attributes encoded numerically
- Contains missing values represented by “?”
- Requires preprocessing before analysis

6. High Dimensionality Justification

The dataset contains multiple medical parameters describing cardiovascular health. Many of these clinical measurements are physiologically related and may exhibit correlation with each other.

Such datasets with several interdependent attributes can be considered moderately high-dimensional in medical data analysis. Therefore, dimensionality reduction techniques such as Principal Component Analysis (PCA) are useful for extracting dominant health-related patterns and reducing redundancy among features.



7. Preprocessing Requirements

Before applying machine learning techniques, the following preprocessing steps are required:

- Replace missing values represented by “?”
- Remove incomplete records
- Convert all attributes to numeric format
- Convert target into binary classification (disease / no disease)

These steps ensure the dataset becomes suitable for statistical analysis and PCA transformation.